Communities of the Interested

Redistricting Institution Congruence with Communities of Interest Among Geocoded Public Comments

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0.1 Redistricting and Congruence

Existing literature has found that state legislatures are generally responsive to public opinion but not congruent with the concrete policies desired by public majorities (Lax and Phillips 2012). Like many other policy issue areas, the clandestine practice of redistricting is not immune to this congruency gap. Partisan legislature redistricting outcomes often produce large "efficiency gaps" between statewide vote shares and resulting seat counts for each political party (Warshaw, McGhee, and Migurski 2022). Redistricting, especially in recent cycles, has also widely reduced the frequency of both competitive electoral districts (Jasny et al. 2025; Kenny et al. 2023) and majority-minority or majority-influence districts (Barabas and Jerit 2004). However, unlike most issue areas, redistricting decisions produce categorical and geographically-defined results on the same dimensional plane as any hypothetical, concrete public preference about redistricting. Instead of trying to evaluate how the specific language of a piece of legislation translates onto individual policy preferences, which can be untethered by practical and semantic constraints, redistricting decisions present an opportunity for both policy and mass opinion can be mapped onto sets of precincts in a given state with measurable degrees of overlap. This makes redistricting a powerful arena for the study of congruence more broadly, especially with redistricting being so closely linked to the health of America's state and federal democracies (Grumbach 2023; Gelman and King 1994; Fraga 2016).

Measuring congruence with redistricting decisions, however, requires that we understand what most members of the public would request of their boundary-drawing institutions in the first place. This is no easy task. Public knowledge and opinions on redistricting, as a concept, are relatively sparse (Cooper et al. 2024; McCarthy 2022), with respondents often relying on partisan cues or judgements about the

overall redistricting process. Evaluating public opinion on redistricting is further complicated by low levels of geographic literacy across the country (McFarlane 2024), meaning that citizens are unlikely to have the expertise to properly judge an entire state's district map. True opinions on a map among the public are likely localized and related to the partisan or nonpartisan factors of the communities Americans call home. A citizen might not have a good sense of how to divide up an entire metro area into state house districts, but they can probably comment on whether their neighborhood, city, school district, and county should be kept whole, combined with other areas, or separated from other areas. Place based identities are a powerful force in the United States (Elder and Lueders 2025), implying that residents of a community have some perception of the social, cultural, economic, and geographic dynamics that separate "where they live" from everywhere else. It is under these conditions that a key redistricting criteria - the concept of "Communities of Interest" - can offer a critical advantage over other outcome heuristics in the translation of public opinion to geographic configurations that enacted maps can either be congruent or incongruent with.

0.2 Communities of Interest in Legislative Redistricting

Nicholas Stephanopoulos (2012) articulates that a group of people constitute a community of interest (COI) if they (1) share a common geographic range, (2) share common socioeconomic or cultural interests, and (3) subjectively identify as members of a cohesive geographic community. Respecting communities of interest is typically one of several criteria that many boundary-drawing institutions are constitutionally obligated to respect (Webster 2013), including contiguity, compactness, population equality, and protection of minority voting populations. However, these criteria are difficult to balance, and some can even be contradictory. In fact, the "reapportionment revolution" of the 1960s, a series of Supreme Court decisions requiring that legislative redistricting strictly adhere to the criteria of population equality across districts, is credited with massively disrupting the ability of states to respect COIs despite the clear benefits of meeting both criteria simultaneously (Stephanopoulos 2012). State and federal courts are nonetheless receptive of arguments in favor of protecting COIs, often striking down maps in clear violation of the criteria. Furthermore, recent court decisions have diminished the prominence of the Voting Rights Act as a basis for striking down statewide maps that fail to appropriately represent minority populations (Chen et al. 2022), leaving

communities of interest as a possible line of defense for these populations to still litigate with.

Thus, a burgeoning community of scholars, recognizing the utility offered by communities of interest as well as their importance in district plan litigation, have articulated a variety of methods for COI measurement. Some scholars have used contextual and behavioral characteristics of smaller geographic units within states (Kruse, Gao, and Mayer 2026; Gimpel and Harbridge-Yong 2020; Makse 2012; Syme 2025; Rossiter, Wong, and Delamater 2018; Swanson and Bryan 2024), clustering units into cohesive groups and comparing district maps to group boundaries. For example, Kruse, Gao, and Mayer (2026) use Facebook friendship networks and mobile device visits between Census block groups to construct COIs, while Maske sorts communities based on issue area initiative votes in Ohio (Makse 2012). These observational approaches present defensible solutions based on well-distributed data across entire states, and their sophisticated methods have yielded actionable insights, but they nonetheless ascribe community membership at a distance from the public's participation in the redistricting process.

Other scholars, therefore, have constructed COIs based on grassroots input from the public, including the deployment of public mapping tools that allow respondents to "draw" their communities of interest from a bird's eye view (Chen et al. 2022; Chambers et al. 2022). These drawn communities of interest can then be compiled and analyzed using a variety of computational techniques. Chambers and colleagues cluster overlapping COIs based on their similarity across respondents, while Chen and colleagues introduce novel measures to evaluate how egregiously an enacted plan splits a given community. These studies provide a framework that boundary-drawing institutions can use to engage the public and better link redistricting with democratic participation. They also provide a closer estimation of the kind of communication flows between constituents and legislators that are likely to occur during the redistricting process.

The most widespread attempts at measuring communities of interest, however, have thankfully been made the states themselves, with boundary-drawing institutions launching widespread outreach programs in recent redistricting cycles. Many institutions offer public hearings so that concerned citizens can deliver speeches and presentations to boundary drawers, and lengthy essay submissions or even complete district maps drawn using publicly available GIS software are common forms of participation. But the largest body of testimony collected in the most recent redistricting cycle came from state-run public web comment platforms,

with submissions numbering in the hundreds or thousands for many states, especially those making use of an independent redistricting committee. Among these comments, communities of interest discussion often naturally takes center stage, assembling a rich mosaic of geographic configuration requests that span entire states. As such, these web comments provide the strongest link to date connecting politically engaged residents to the redistricting process, but questions remain regarding how well these comments represent the public and whether any requests are congruent with enacted maps.

Not only are these recent testimony gathering efforts widespread, but they also reflect, as with many other issue areas, the data that legislators themselves collect to gather opinions from constituents and make subsequent decisions, making public web comments ideally situated for answering several questions about if and why congruence lags during the redistricting process. Are public commenters independent testifiers resembling a cross section of their communities, or do they form unrepresentative, coordinated networks of advocacy across space and time? Are redistricting institutions responsive to requests of a certain nature or popularity? Across states, what types of redistricting institutions are the most responsive to community of interest testimony? Answers to each of these questions can lend insight to the broader study of public testimony, responsiveness, congruence, and democratic representation.

0.3 Theories of Congruence with Public Comment

Gabriella Subia's exploration of public participation during the creation of Colorado's eighth congressional district suggests that the state's IRC prioritized the representation of Hispanic voters, a competitive partisan balance, and the protection of various industries north of Denver, all of which were advocated for by both interest groups and local residents (Subia 2024). This case study aligns itself with the broader literature on public comment during rulemaking, which has generally found that institutions are responsive to those who participate (Sahn 2025; Ingrams 2023). One source of district map incongruence, then, might derive from the fact that those who testify have their requests met, but they don't appropriately represent a state's constituency, leading to the following hypothesis:

H1: Across individual states, public web comment density and commenter demographics will not

be representative of local community sizes and contexts.

Another analysis by Olivia Rambo finds among a sample of comments submitted to the California Citizen's Redistricting Commission that certain interests were more likely to achieve their boundary drawing aims than others, and many comments were copies of templates distributed to the public by interest groups (Rambo 2021). This study is consistent with the idea that even among those who participate, the most well-funded and well-coordinated voices during the process of collecting public testimony have the greatest influence (Yackee and Yackee 2006), highlighting a second source of disconnect between mass opinion and redistricting outcomes. These organized interests can recruit or inspire bundles of comments making similar requests from similar areas at similar times, either using templates (Rambo 2021) or encouraging commenters to express a request in their own words.

Furthermore, for an issue like redistricting, an issue that the public has a limited knowledge of (Cooper et al. 2024), interest groups can either advocate for policy that appropriately represents the interests of their corresponding communities, or they can try to pull a district plan farther away from the will of underrepresented communities. Other comments will appear organically among the public, with altruistic, independent commenters - whether they be active community members, academics, or political hobbyists - genuinely expressing the best interests of a local population. Under the relative anonymity of many web commenting platforms, a boundary-drawing institution might struggle to discern the motivations behind popular requests among the comments they receive. Furthermore, boundary-drawing institutions can defend their decisions against the threat of litigation or veto by claiming they're responsive to the input they received, arguing that the most outspoken voices during the redistricting process are well informed about the appropriate decisions for their communities more broadly. Thus, we test the following two hypotheses about the nature of public web comment corpora:

H2: Across individual states, public web comments will not be independent, and commenters will span a network of advocacy detectable by clusters of spaciotemporal comment density.

H3: Within a state, congruence with some subset of public comment requests will be higher than what would be expected in the absence of public comments.

Which requests, contingent on support of H2 and H3, should we expect to be more congruent than others? A single state's redistricting institution has a collective set of goals and a necessarily unique approach to synthesizing any public opinion they receive. Previous research has found that legislators will often disregard public input that they disagree with (Butler and Dynes 2016), and boundary drawers must also balance a variety of other redistricting criteria besides adherence to communities of interest (Webster 2013). Thankfully, McCartan and colleagues have built measures based on a Nash equilibrium process to assess the extent that partisan actors have leeway over the creation of final maps - based on initial boundary drawing responsibilities, veto guardrails, and the likelihood of court challenges - from state to state (2026). Unsurprisingly, states with independent redistricting commissions tend to have the lowest "partisan leeway" scores, while states placing the responsibility solely in the hands of a partisan legislature have much higher scores. States with greater partisan leeway are likely to be choosier about the public community of interest comments they consider, adhering closely to clusters of requests that align with their broader agenda while completely discounting others. It may also be the case that systematic differences in congruence emerge based on the racial or geographic makeup of a request's commenters or the nature of the request itself. This leads to the following hypotheses:

H4: The association between request congruence and request popularity will be less predictable among states with greater partisan leeway over the redistricting process.

H5: The association between request congruence and compositional features of a request's commenters will be stronger among states with greater partisan leeway over the redistricting process.

Support for each of these hypotheses can inform how congruence is disconnected from public opinion across many other issue areas (Lax and Phillips 2012). Unlike these other issues, however, the geographic results of redistricting enable direct measurements of congruence with constituent requests, and the criteria of communities of interest serves as a vital link between credible mass opinion and actionable geographic configurations. Exploring the nature of testimony that institutions collect and respond to can inform how statehouse democracies, some more than others, unintentionally or deliberately misalign themselves with the will of the people.

References

Barabas, Jason, and Jennifer Jerit. 2004. "Redistricting Principles and Racial Representation." *State Politics & Policy Quarterly* 4 (4): 415–35. https://doi.org/http://www.jstor.org/stable/40421529.

Butler, Daniel, and Adam Dynes. 2016. "How Politicians Discount the Opinions of Constituents with Whom They Disagree." *American Journal of Political Science* 60 (4): 975–89.

Chambers, Erin, Moon Duchin, Ranthony Edmonds, Parker Edwards, J N Matthews, Anthony Pizzimenti, Chanel Richardson, Parker Rule, and Ari Stern. 2022. "Aggregating Community Maps." *Proceedings of the 30th International Conference on Advances in Geographic Information Systems*, no. 27. https://doi.org/10.1145/3557915.3560961.

Chen, Sandra, Samuel Wang, Bernard Grofman, Richard Oberand, Kyle Barnes, and Jonathan Cervas. 2022. "Turning Communities of Interest into a Rigorous Standard for Fair Districting." *Stanford Journal of Civil Rights and Civil Liberties* 18 (1): 101–89.

Cooper, Christopher, Scott Huffmon, H. Gibbs Knotts, and Seth McKee. 2024. "Partisanship, Race, and Public Opinion on Redistricting: The Case of South Carolina." *Election Law Journal: Rules, Politics, and Policy* 23 (4): 333–44. https://doi.org/10.1089/elj.2023.0048.

Elder, Elizabeth, and Hans Lueders. 2025. "Place Attachments: Theory and Measurement for Political Science." *Political Behavior*. https://doi.org/10.1007/s11109-025-10076-w.

Fraga, Bernard. 2016. "Redistricting and the Causal Impact of Race on Voter Turnout." *The Journal of Politics* 78 (1): 19–34. https://doi.org/10.1086/683601.

Gelman, Andrew, and Gary King. 1994. "Enhancing Democracy Through Legislative Redistricting." *American Political Science Review* 88 (3): 541–59. https://doi.org/https://www.jstor.org/stable/2944794.

Gimpel, James, and Laurel Harbridge-Yong. 2020. "Conflicting Goals of Redistricting: Do Districts That Maximize Competition Reckon with Communities of Interest?" *Election Law Journal: Rules, Politics, and Policy* 19 (4): 451–71. https://doi.org/https://doi.org/10.1089/elj.2019.0576.

Grumbach, Jacob. 2023. "Laboratories of Democratic Backsliding." *American Political Science Review* 117 (3): 967–84. https://doi.org/10.1017/S0003055422000934.

Ingrams, Alex. 2023. "Do Public Comments Make a Difference in Open Rulemaking? Insights from Infor-

mation Management Using Machine Learning and QCA Analysis." *Government Information Quarterly* 40 (1): 101778. https://doi.org/https://doi.org/10.1016/j.giq.2022.101778.

Jasny, Ethan, Christopher Kenny, Cory McCartan, Tyler Simko, Melissa Wu, Michael Zhao, Aneetej Arora, et al. 2025. "Gerrymandering and Geographic Polarization Have Reduced Electoral Competition." *arXiv*. https://doi.org/https://arxiv.org/abs/2508.15885.

Kenny, Christopher, Cory McCartan, Tyler Simko, Shiro Kuriwaki, and Kosuke Imai. 2023. "Widespread Partisan Gerrymandering Mostly Cancels Nationally, but Reduces Electoral Competition." *Proceedings of the National Academy of Sciences* 120 (25): e2217322120. https://doi.org/10.1073/pnas.2217322120.

Kruse, Jacob, Song Gao, and Kenneth Mayer. 2026. "Modeling Region Affiliation with Fuzzy Membership Based on Spatial and Social Interactions." *Annals of the American Association of Geographers* 116 (2): 289–311. https://doi.org/10.1080/24694452.2025.2551044.

Lax, Jeffrey, and Justin Phillips. 2012. "The Democratic Deficit in the States." *American Journal of Political Science* 56 (1): 148–66. https://doi.org/http://www.jstor.org/stable/23075149.

Makse, Todd. 2012. "Defining Communities of Interest in Redistricting Through Initiative Voting." *Election Law Journal: Rules, Politics, and Policy* 11 (4): 503–17. https://doi.org/https://doi.org/10.1089/elj.2011.0144.

McCartan, Cory, Christopher Kenny, Tyler Simko, Emma Ebowe, Michael Zhao, and Kosuke Imai. 2026. "Redistricting Reforms Reduce Gerrymandering by Constraining Partisan Actors." *American Political Science Review*, 1–20. https://doi.org/10.1017/S0003055426101610.

McCarthy, Devin. 2022. "Limits of Partisanship in Citizen Preferences on Redistricting." *Election Law Journal: Rules, Politics, and Policy* 21 (2): 155–70. https://doi.org/10.1089/elj.2020.0709.

McFarlane, Donovan. 2024. "Critical Review of Geographic Illiteracy in the United States: A Multifaceted Analysis." *Journal of Geography, Environment and Earth Science International* 28 (1): 11–25. https://doi.org/10.9734/jgeesi/2024/v28i1741.

Rambo, Olivia. 2021. "Defining Communities of Interest: Political Self-Definition in Los Angeles County During the First California Citizen's Redistricting Commission." *GeoJournal* 86: 1545–87. https://doi.org/https://doi.org/10.1007/s10708-020-10141-3.

Rossiter, Kalyn, David Wong, and Paul Delamater. 2018. "Congressional Redistricting: Keeping Communities Together?" *The Professional Geographer* 70 (4): 609–23. https://doi.org/https://doi.org/10.1080/00330124.2018.1443477.

Sahn, Alexander. 2025. "Public Comment and Public Policy." *American Journal of Political Science* 69 (2): 685–700. https://doi.org/https://doi.org/10.1111/ajps.12900.

Stephanopoulos, Nicholas. 2012. "Redistricting and the Territorial Community." *University of Pennsylvania Law Review* 160 (5): 1379.

Subia, Gabriella. 2024. "Creating Colorado's 8th Congressional District." *Journal of Maps* 20 (1): 2415644. https://doi.org/10.1080/17445647.2024.2415644.

Swanson, David, and T. M. Bryan. 2024. "Using Cluster Analysis to Identify Communities of Interest for Purposes of Legislative Redistricting: A Case Study of Parishes in Louisiana." *Papers in Applied Geography* 10 (3): 217–31. https://doi.org/https://doi.org/10.1080/23754931.2024.2346326.

Syme, James. 2025. "Using Dual Graphs to Model Territorial Communities of Interest." *PREPRINT*. https://doi.org/https://doi.org/10.21203/rs.3.rs-7040797/v1.

Warshaw, Christopher, Eric McGhee, and Michal Migurski. 2022. "Districts for a New Decade—Partisan Outcomes and Racial Representation in the 2021–22 Redistricting Cycle." *Publius: The Journal of Federalism* 52 (3): 428–51. https://doi.org/10.1093/publius/pjac020.

Webster, Gerald. 2013. "Reflections on Current Criteria to Evaluate Redistricting Plans." *Political Geography* 32: 3–14. https://doi.org/https://doi.org/10.1016/j.polgeo.2012.10.004.

Yackee, Jason Webb, and Susan Webb Yackee. 2006. "A Bias Towards Business? Assessing Interest Group Influence on the u.s. Bureaucracy." *The Journal of Politics* 68 (1): 128–39. https://doi.org/https://doi.org/10.1111/j.1468-2508.2006.00375.x.